

Bahar Partov Poor

baharxy.github.io

Researcher & Technologist · Computer Science PhD

baharp@csail.mit.edu

GitHub 1 GitHub 2

LinkedIn

RESEARCH PROFILE

Research Interests: Optimization and decision-making under uncertainty; energy-efficient, responsible, and secure AI systems; and exploring internet security boundaries.

Research Output: 14 peer-reviewed publications (conference and journal) with 188 citations. Full list available at [Google Scholar](#).

Technical Skills: Network programming, network simulation, machine learning, distributed algorithms, pentesting, CI/CD pipelines, agentic AI systems, optoelectronic devices, programmable radios, IoT sensors. Languages: Python, C++, Matlab, Go, Perl, Bash, R, Java.

Languages: English (fluent) · French (B2) · Persian (native).

EDUCATION

•PhD in Computer Science

08/2012 – 09/2015

Thesis: Resource Allocation for Next Generation RANs

Hamilton Institute / Nokia Bell Labs, Dublin, Ireland

- Advisor: Prof. Douglas Leith. Committee: Prof. Merouane Debbah, Dr. Rudi Villing, Dr. Catherine Hurley.
- Industrial advisors: Dr. Holger Claussen & Dr. Rouzbeh Razavi (Bell Labs).
- Funded by a full scholarship from the Hamilton Institute and Nokia Bell Labs.

•MSc in Telecommunications and Information Systems

09/2008 – 11/2009

Thesis: Novel Video Streaming over IEEE 802.16 Networks

University of Essex, UK

- GPA: 87/100. Graduated with Distinction. Awarded the **Telecom Technologies Prize** for outstanding performance.
- Advisors: Prof. Mohammed Ghanbari, Prof. Martin Fleury.

•BSc in Electrical Engineering (minor: Communications Engineering)

09/2003 – 10/2007

Ranked 1st among communications engineering students.

Iran

- GPA: 17.53/20.

RESEARCH EXPERIENCE

•Visiting Postdoctoral Fellow

09/2016 – 03/2017

CSAIL (Katabi Lab) & Sensable City Lab

MIT, Cambridge, USA

- Contributed to the design and initiation of new projects at the Sensable City Lab, including the Philips smart lighting and city scanner projects (Python, Bash, Matlab).
- Developed synchronisation algorithms for large-scale LTE networks (C++, srsLTE) at Katabi's Lab, achieving a **2.7× throughput improvement** over conventional methods.

•Research Fellow

11/2015 – 09/2016

School of Computer Science and Statistics (NetLab)

Trinity College Dublin, Dublin, Ireland

- Designed and conducted experiments on the impact of network delay on LTE and WiFi links (Python, Bash), generating insights for future multipath TCP scheduler design.
- Built lightweight, privacy-preserving recommendation systems for WiFi AP selection using supervised learning (Matlab).
- Investigated distributed optimisation methods for network utility maximisation and auction algorithms.

•PhD Research Fellow

08/2012 – 08/2015

Hamilton Institute, Maynooth University

Nokia Bell Labs, Dublin, Ireland

- Proposed mathematical models for resource allocation in 5G radio access networks using convex optimisation and game theory.
- Applied machine learning, data analytics, and distributed optimisation to wireless resource management problems (Matlab, R, Java).
- Produced **6 peer-reviewed publications** during the PhD (part of 14 total career publications with **188 citations**).

TEACHING & SUPERVISION

•Lecturer and Mentor

Delivered workshops and mentored selected students nationwide

06/2018, 06/2019
AI for Social Good Summer School, Canada

•Graduate Student Supervisor

Supervised internships funded by NSERC and Mitacs grants (Dalhousie, Polytechnique Montréal)

2017 – 2018
Montréal, Canada

•Teaching Assistant — STU33009: Statistical Methods for Computer Science

BSc Computer Science (Year 3). Supported students in weekly tutorials covering probability, random variables, statistical inference, and machine learning

•Seminar Organiser

Co-ordinated weekly research seminar series at the School of Computer Science & Statistics Trinity College Dublin, Dublin, Ireland

GRANTS & FUNDING

•25K CAD — NSERC Canada

Design of integrated circuits for ultra low-power RF energy detection.

08/2018
with Dalhousie University

•60K CAD — Mitacs Canada (two grants)

Graduate student internship opportunities.

05/2017, 10/2017
University-industrial partnership

•25K CAD — NSERC Canada

Graduate internship: evaluation of low-power digital design techniques.

11/2017
with Polytechnique Montréal

•600K CAD Investment Fund

Seed funding for Wavelite / Haila Technologies.

07/2017
TandemLaunch Inc.

•Travel Grant

10/2016
Computing Research Association / ACM SIGMOBILE (MobiCom)

•N2 Women Young Researcher Fellowship

02/2015
Supported by Microsoft Research & HP Labs

•Full PhD Scholarship

08/2012 – 08/2015
Hamilton Institute & Nokia Bell Labs

INDUSTRY EXPERIENCE

•Engineering Manager, Security

Tucows Inc

05/2025 – present

Remote

– Leading the engineering team and cross-functional SDLC security initiatives, while remaining hands-on with back-end security and DNS abuse mitigation.

– Applying AI-driven tooling to strengthen security operations and SDLC workflows.

•Software Engineer: Back-end & Security

Tucows Inc

11/2020 – 04/2025

Remote

– Developed security plugins and abuse-detection packages for large-scale networked systems (Go, Perl, Lua, Bash); designed REST APIs and micro-service observability platform (Grafana, PostgreSQL, RabbitMQ).

– Represented Tucows in ICANN's CPH/TechOps working group.

•Co-Founder & Tech Lead

Wavelite / Haila Technologies Inc, incubated at TandemLaunch VC

03/2017 – 05/2019

Montréal, Canada

– Raised 600K CAD; built passive IoT sensor prototypes using WiFi signals (C++, FPGAs, MQTT).

– Received CES Eureka Park Climate Change Award (2018); covered by RFID Journal; signed development contract with a major German home automation company.

•Engineer

Oclaro Inc (now Lumentum)

02/2010 – 08/2012

Paignton, UK

– Developed test automation (C#) and contributed to applied research on optical lasers for wind turbine sensing.

SCIENTIFIC SERVICE & COMMUNITY

•Reviewer & Programme Committee Member

Leading venues in wireless networking and communications

11/2012 – 06/2019

IEEE Conferences and Journals

•Team Lead & Co-Founder

Organised career fairs and monthly professional events

05/2018 – 10/2020

Women of MENA in Tech, Montréal

•Volunteer

Neighbourhood social support services

06/2022 – present

Service d'Entraide Saint-Romuald

•Volunteer Mapper

Crowd-mapping rural Tanzania to support anti-FGM efforts

01/2018 – 11/2018

Crowd2map Tanzania